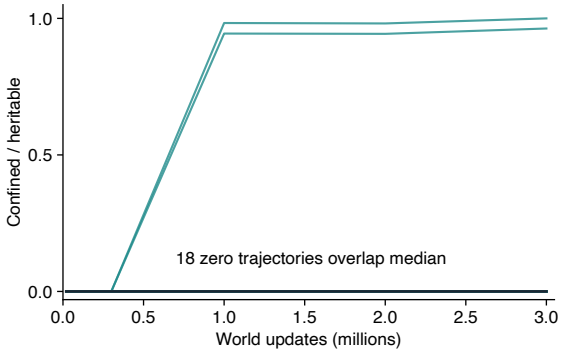
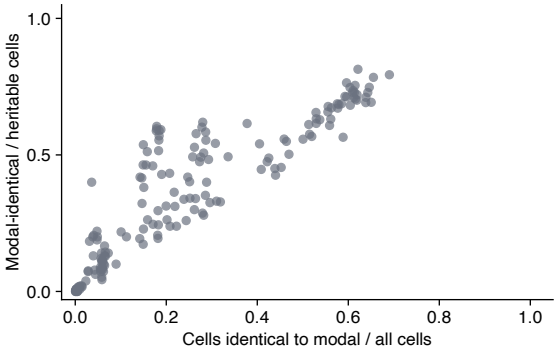
**a** **Random-register reset: Z80 L16, 20 worlds**




At 3 million updates: 2/20 worlds majority confined
Thin = worlds; black median; cells confined / heritable cells
Confined: no partner instruction-stream fetch in 16/16 traces

**b** **Random cells versus the modal tape**



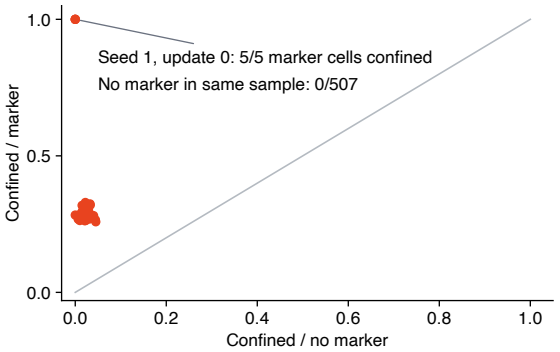

One dot = final world sample; 170 worlds in cohorts of c
Y: sampled heritable cells identical to modal / sampled heritable cells

**c** **Class denominators by cohort**

| Cohort | Worlds | Cell rows | Confined rows | T rows |
|---|---|---|---|---|
| G, L16 | 20 | 3938 | 2624 | 259 |
| G, L20 | 20 | 3757 | 2032 | 358 |
| G, L50 | 20 | 5152 | 1165 | 19 |
| G, L64 | 20 | 5444 | 897 | 320 |
| I, L16 | 10 | 1217 | 1217 | 1008 |
| K, L32 | 20 | 5519 | 741 | 149 |
| L, L16 | 10 | 3210 | 2649 | 112 |
| L, L20 | 10 | 2556 | 1783 | 253 |
| M, L16 | 20 | 3966 | 2536 | 0 |
| M, L32 | 20 | 5600 | 11 | 0 |

Pooled sampled-cell rows; confined and T are subsets
T = confined with ≥5 scored sites
G benign Z80; I lethal Z80; K: L32 ring aligned with 16-bit address wrap
L: closure-recovery extensions to 10 million updates
M 8080 subset; L in labels = tape bytes (not cohort L)

**d** **Sequence marker is not confinement**




Z80 L50 matched-pair invasion: 3 worlds, 24 samples
Sequence marker: 20 f0 (hex bytes); positive/negative cells
512 cells/sample; equal-fraction diagonal

**e** **Exploratory detection order: any detection**

| Cohort | World count | R first | Same update | T first | R only | T only | None |
|---|---|---|---|---|---|---|---|
| G, L16 | 20 | 0 | 1 | 1 | 17 | 1 | 0 |
| G, L20 | 20 | 4 | 9 | 0 | 3 | 3 | 1 |
| G, L50 | 20 | 8 | 1 | 0 | 11 | 0 | 0 |
| G, L64 | 20 | 18 | 0 | 0 | 2 | 0 | 0 |
| I, L16 | 10 | 1 | 1 | 5 | 0 | 3 | 0 |
| K, L32 | 20 | 1 | 8 | 2 | 5 | 0 | 4 |
| L, L16 | 10 | 0 | 0 | 1 | 9 | 0 | 0 |
| L, L20 | 10 | 4 | 4 | 2 | 0 | 0 | 0 |
| M, L16 | 20 | 0 | 0 | 0 | 20 | 0 | 0 |
| M, L32 | 20 | 0 | 0 | 0 | 8 | 0 | 12 |

R = confined ≤2 scored sites; T = confined ≥5 scored sites
R<T: R observed first; Same: same sampled update; T<R: T first
Any = positive estimated class share; n = worlds; not ancestry

**f** **Detection-order threshold sensitivity**

| Threshold | R first | Same update | T first | Both seen | World count |
|---|---|---|---|---|---|
| G, L16 >50% | 0 | 0 | 0 | 0 | 20 |
| G, L16 >10% | 0 | 0 | 1 | 1 | 20 |
| G, L20 >50% | 0 | 0 | 2 | 2 | 20 |
| G, L20 >10% | 6 | 5 | 0 | 11 | 20 |
| G, L50 >50% | 0 | 0 | 0 | 0 | 20 |
| G, L50 >10% | 1 | 0 | 0 | 1 | 20 |
| G, L64 >50% | 2 | 0 | 2 | 4 | 20 |
| G, L64 >10% | 8 | 0 | 0 | 8 | 20 |
| I, L16 >50% | 1 | 0 | 0 | 1 | 10 |
| I, L16 >10% | 1 | 1 | 1 | 3 | 10 |
| K, L32 >50% | 0 | 0 | 1 | 1 | 20 |
| K, L32 >10% | 1 | 5 | 3 | 9 | 20 |
| L, L16 >50% | 0 | 0 | 1 | 1 | 10 |
| L, L16 >10% | 0 | 0 | 1 | 1 | 10 |
| L, L20 >50% | 0 | 0 | 2 | 2 | 10 |
| L, L20 >10% | 5 | 3 | 2 | 10 | 10 |
| M, L16 >50% | 0 | 0 | 0 | 0 | 20 |
| M, L16 >10% | 0 | 0 | 0 | 0 | 20 |
| M, L32 >50% | 0 | 0 | 0 | 0 | 20 |
| M, L32 >10% | 0 | 0 | 0 | 0 | 20 |

R = confined ≤2 scored sites; T = confined ≥5 scored sites
Threshold: R or T cells / heritable cells
Both detected = sum of the three order columns; n = worlds
Site: ≥half tested byte values heritable, allele in ≥50% qualifying copies